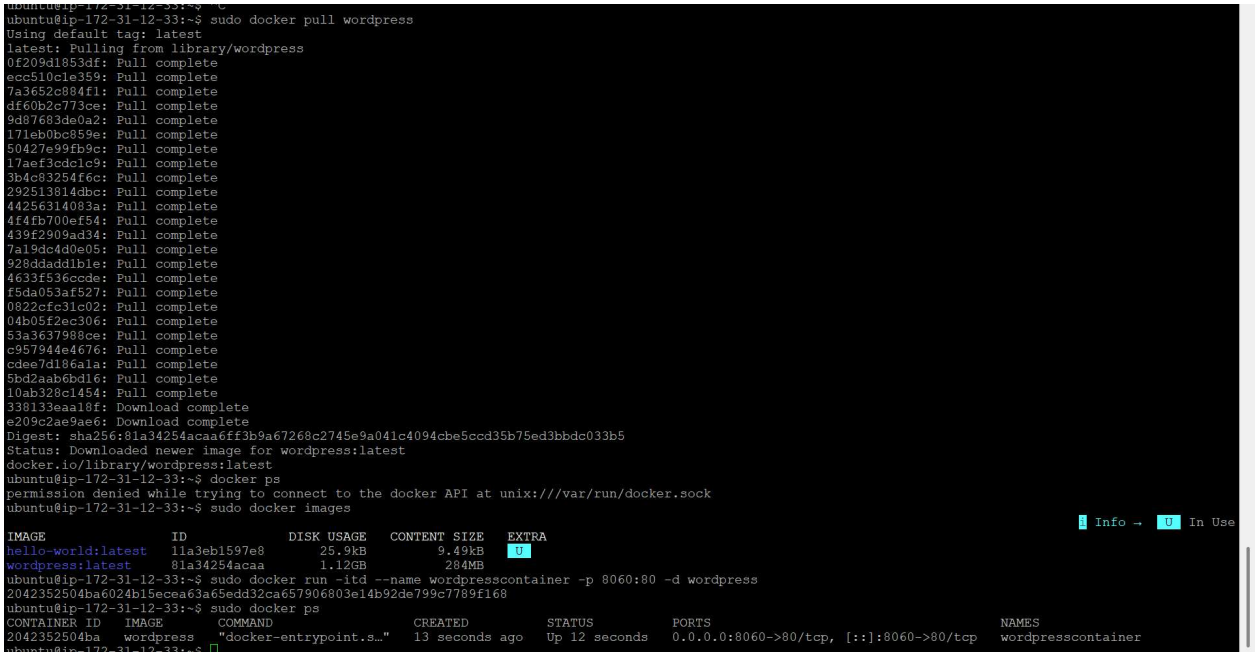
Task1 - Linux 10 commands

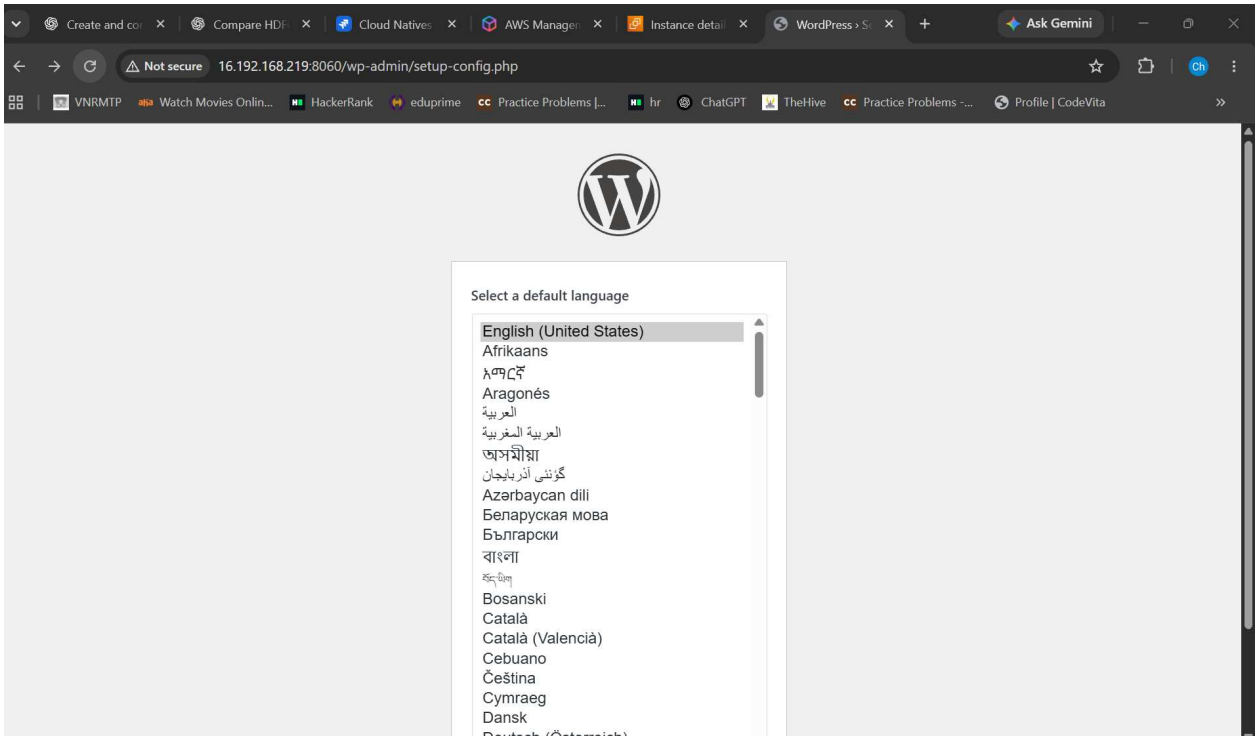
```
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ ps
    PID TTY          TIME CMD
    347 pts/0    00:00:00 bash
    503 pts/0    00:00:00 ps
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  9 09:30 ?        00:00:00 /sbin/init
root           2       1  0 09:30 hvc0     00:00:00 /init
root           7       2  0 09:30 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4
root          50       1  2 09:30 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+      82       1  1 09:30 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          90       1  3 09:30 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root         115       1  0 09:30 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chron
root         121       1  0 09:30 ?        00:00:00 /usr/sbin/cron -f -P
message+     125       1  0 09:30 ?        00:00:00 @dbus-daemon --system --address=system
root         165       1  2 09:30 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dis
root         171       1  3 09:30 ?        00:00:00 /usr/lib/snapd/snapd
root         173       1  1 09:30 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         180       1  1 09:30 ?        00:00:00 /usr/libexec/wsl-pro-service
syslog       192       1  0 09:30 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         229       1  0 09:30 tty1     00:00:00 /usr/sbin/agetty --noreset --noclear -
_chrony      248     115  0 09:30 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         253       1  2 09:30 ?        00:00:00 /usr/bin/python3 /usr/share/unattended
_chrony      259     248  0 09:30 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         295       1  1 09:30 ?        00:00:00 /usr/lib/systemd/systemd-timedated
root         345       2  0 09:30 ?        00:00:00 /init
root         346     345  0 09:30 ?        00:00:00 /init
sharant+     347     346  1 09:30 pts/0    00:00:00 -bash
root         348       2  0 09:30 ?        00:00:00 login -- sharanteja2211
sharant+     393       1  2 09:30 ?        00:00:00 /usr/lib/systemd/systemd --user
sharant+     395     393  0 09:30 ?        00:00:00 (sd-pam)
sharant+     427     348  0 09:30 pts/1    00:00:00 -bash
sharant+     513     347  0 09:30 pts/0    00:00:00 ps -ef
```

```
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ uptime
 09:31:01 up 0 min,  1 user,  load average: 0.14, 0.05, 0.02
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ free -h
               total        used        free      shared  buff/cache   available
Mem:            11Gi       617Mi        10Gi       3.5Mi       664Mi        10Gi
Swap:          3.0Gi          0B       3.0Gi
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ ps -ef | head -n 1-
head: invalid number of lines: '1-'
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ ps -ef | head -n 10
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  1 09:30 ?        00:00:00 /sbin/init
root           2       1  0 09:30 hvc0     00:00:00 /init
root           7       2  0 09:30 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          50       1  0 09:30 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+      82       1  0 09:30 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          90       1  0 09:30 ?        00:00:00 /usr/lib/systemd/systemd-udevd
root         115       1  0 09:30 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         121       1  0 09:30 ?        00:00:00 /usr/sbin/cron -f -P
message+     125       1  0 09:30 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activatio
n --syslog-only
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ sleep 300 &
[1] 587
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ jobs
[1]+  Running                 sleep 300 &
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ pgrep  -af "sleep 300"
587 sleep 300
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ kill -TERM 587
[1]+  Terminated              sleep 300
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ jobs
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ sleep 200
^Z
[1]+  Stopped                 sleep 200
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ jobs
[1]+  Stopped                 sleep 200
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ bg
[1]+ sleep 200 &
```

```
sharanteja2211@SharanTeja:/mnt/c/windows/system32$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.1  24140 15388 ?        Ss   09:30   0:01 /sbin/init
root           2  0.0  0.0   3180  2204 hvc0     Sl+  09:30   0:00 /init
root           7  0.0  0.0   3180  2040 hvc0     Sl+  09:30   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
root          50  0.0  0.1  50396 16872 ?        S<s  09:30   0:00 /usr/lib/systemd/systemd-journald
systemd+      82  0.0  0.1  22416 14456 ?        Ss   09:30   0:00 /usr/lib/systemd/systemd-resolved
root          90  0.0  0.1  35484 12424 ?        Ss   09:30   0:00 /usr/lib/systemd/systemd-udevd
root         115  0.0  0.0   2888  2004 ?        Ss   09:30   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         121  0.0  0.0   4472  3132 ?        Ss   09:30   0:00 /usr/sbin/cron -f -P
message+     125  0.0  0.0   8800  5428 ?        Ss   09:30   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
root         165  0.0  0.2  44096 29756 ?        Ss   09:30   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-trigge
root         173  0.0  0.0  18440  9468 ?        Ss   09:30   0:00 /usr/lib/systemd/systemd-logind
syslog       192  0.0  0.0 220548  5936 ?        Ssl  09:30   0:00 /usr/sbin/rsyslogd -n -iNONE
root         229  0.0  0.0   5492  2812 tty1     Ss+  09:30   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/
_chrony      248  0.0  0.0  20424  6380 ?        S    09:30   0:00 /usr/sbin/chronyd -n -F 1 -x
root         253  0.0  0.2 123460 32608 ?        Ssl  09:30   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
_chrony      259  0.0  0.0  12096  2532 ?        S    09:30   0:00 /usr/sbin/chronyd -n -F 1 -x
root         345  0.0  0.0   3196  1112 ?        Ss   09:30   0:00 /init
root         346  0.0  0.0   3196  1248 ?        S    09:30   0:00 /init
sharant+     347  0.0  0.0   6784  6116 pts/0    Ss   09:30   0:00 -bash
root         348  0.0  0.0   8648  5604 ?        Ss   09:30   0:00 login -- sharanteja2211
sharant+     393  0.0  0.1  22344 13616 ?        Ss   09:30   0:00 /usr/lib/systemd/systemd --user
sharant+     395  0.0  0.0  22920  4100 ?        S    09:30   0:00 (sd-pam)
sharant+     427  0.0  0.0   6136  5476 pts/1    Ss+  09:30   0:00 -bash
sharant+     834 33.3  0.0   7160  4356 pts/0    R+   09:43   0:00 ps aux
```

Task 2 - pull docker images wordpress mysql





mysql

```
CONTAINER ID   IMAGE         COMMAND                  CREATED              STATUS                      PORTS                                       NAMES
b7101f3e9e33   mysql:8.0     "docker-entrypoint.s…"   About a minute ago   Up About a minute           3306/tcp, 33060/tcp                         mysql-db
2042352504ba   wordpress     "docker-entrypoint.s…"   13 minutes ago       Up 13 minutes               0.0.0.0:8060->80/tcp, [::]:8060->80/tcp   wordpresscon
tainer
e4720bc775b7   hello-world   "/hello"                 19 minutes ago       Exited (0) 19 minutes ago                                               thirsty_grot
hendieck
ubuntu@ip-172-31-12-33:~$ sudo docker exec mysql-db mysqladmin ping -uroot -p'RootLab_2026_Strong!'
mysqladmin: [Warning] Using a password on the command line interface can be insecure.
mysqld is alive
ubuntu@ip-172-31-12-33:~$ sudo docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED          STATUS                      PORTS                                       NAMES
b7101f3e9e33   mysql:8.0     "docker-entrypoint.s…"   2 minutes ago    Up 2 minutes                3306/tcp, 33060/tcp                         mysql-db
2042352504ba   wordpress     "docker-entrypoint.s…"   15 minutes ago   Up 15 minutes               0.0.0.0:8060->80/tcp, [::]:8060->80/tcp   wordpresscontain
er
e4720bc775b7   hello-world   "/hello"                 20 minutes ago   Exited (0) 20 minutes ago                                               thirsty_grothend
ieck
ubuntu@ip-172-31-12-33:~$
```

Task3 - Create and Modify a Dockerfile

```
wordpress:latest     81a34254acaa       1.12GB          284MB    U
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker run -itd --name mycontain
er myimage
bb49644673592d16eb1e95f1a0f98a7cf0f82219a305e9f09d2834664b20986d
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED             STATUS
                     PORTS                                       NAMES
bb4964467359   myimage       "/bin/bash"              14 seconds ago      Up 14
seconds                                                          mycontainer
b7101f3e9e33   mysql:8.0     "docker-entrypoint.s…"   57 minutes ago      Up 57
minutes              3306/tcp, 33060/tcp                         mysql-db
2042352504ba   wordpress     "docker-entrypoint.s…"   About an hour ago   Up Abo
ut an hour           0.0.0.0:8060->80/tcp, [::]:8060->80/tcp   wordpresscont
ainer
e4720bc775b7   hello-world   "/hello"                 About an hour ago   Exited
 (0) About an hour ago                                           thirsty_groth
endieck
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker exec -it ^C
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker exec -it bb4964467359 /bi
n/bash
root@bb4964467359:/# vim --version
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
Included patches: 1-16, 647, 678, 697
Modified by team+vim@tracker.debian.org
Compiled by team+vim@tracker.debian.org
Huge version without GUI.  Features included (+) or not (-):
+acl               +file_in_path      +mouse_urxvt       -tag_any_white
+arabic            +find_in_path      +mouse_xterm       -tcl
+autocmd           +float             +multi_byte        +termguicolors
+autochdir         +folding           +multi_lang        +terminal
-autoservername    -footer            -mzscheme          +terminfo
-balloon_eval      +fork()            +netbeans_intg     +termresponse
+balloon_eval_term +gettext           +num64             +textobjects
-browse            -hangul_input      +packages          +textprop
++builtin_terms    +iconv             +path_extra        +timers
+byte_offset       +insert_expand     -perl              +title
+channel           +ipv6              +persistent_undo   -toolbar
+cindent           +job               +popupwin          +user_commands
-clientserver      +jumplist          +postscript        +vartabs
-clipboard         +keymap            +printer           +vertsplit
+cmdline_compl     +lambda            +profile           +vim9script
+cmdline_hist      +langmap           -python            +viminfo
+cmdline_info      +libcall           +python3           +virtualedit
+comments          +linebreak         +quickfix          +visual
+conceal           +lispindent        +reltime           +visualextra
+cryptv            +listcmds          +rightleft         +vreplace
+cscope            +localmap          -ruby              +wildignore
+cursorbind        -lua               +scrollbind        +wildmenu
+cursorshape       +menu              +signs             +windows
```

```
endieck
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker exec -it ^C
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker exec -it bb49644
n/bash
root@bb4964467359:/# vim --version
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
Included patches: 1-16, 647, 678, 697
Modified by team+vim@tracker.debian.org
Compiled by team+vim@tracker.debian.org
```

```
Please send bug reports and questions to <bug-wget@gnu.org>.
root@bb4964467359:/# unzip --version
caution:  both -n and -o specified; ignoring -o
UnZip 6.00 of 20 April 2009, by Debian. Original by Info-ZIP.

Usage: unzip [-Z] [-opts[modifiers]] file[.zip] [list] [-x xlist] [-d exdir]
  Default action is to extract files in list, except those in xlist, to exdir;
  file[.zip] may be a wildcard.  -Z => ZipInfo mode ("unzip -Z" for usage).

  -p  extract files to pipe, no messages     -l  list files (short format)
  -f  freshen existing files, create none    -t  test compressed archive data
  -u  update files, create if necessary      -z  display archive comment only
  -v  list verbosely/show version info       -T  timestamp archive to latest
  -x  exclude files that follow (in xlist)   -d  extract files into exdir
modifiers:
  -n  never overwrite existing files         -q  quiet mode (-qq => quieter)
  -o  overwrite files WITHOUT prompting      -a  auto-convert any text files
  -j  junk paths (do not make directories)   -aa treat ALL files as text
  -U  use escapes for all non-ASCII Unicode  -UU ignore any Unicode fields
  -C  match filenames case-insensitively     -L  make (some) names lowercase
  -X  restore UID/GID info                   -V  retain VMS version numbers
  -K  keep setuid/setgid/tacky permissions   -M  pipe through "more" pager
  -O CHARSET  specify a character encoding for DOS, Windows and OS/2 archives
  -I CHARSET  specify a character encoding for UNIX and other archives
```

```
Please send bug reports and questions to <bug-wget@gnu.org>.
root@bb4964467359:/# wget --version
GNU Wget 1.21.4 built on linux-gnu.

-cares +digest -gpgme +https +ipv6 +iri +large-file -metalink +nls
+ntlm +opie +psl +ssl/openssl

Wgetrc:
    /etc/wgetrc (system)
```

```
othendieck
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker run --rm myima
-n 1"
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
GNU Wget 1.21.4 built on linux-gnu.
bash: line 1: php: command not found
```

```
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker run --rm myimage bash -c "vim --version | head -n 1; wget --version | head -n 1; php --version | head
-n 1"
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
GNU Wget 1.21.4 built on linux-gnu.
PHP 8.3.6 (cli) (built: Sep  2 2026 12:56:02) (NTS)
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker run --rm myimage bash -c "vim --version | head -n 1; wget --version | head -n 1; php --version | head
-n 1"
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
GNU Wget 1.21.4 built on linux-gnu.
PHP 8.3.6 (cli) (built: Sep  2 2026 12:56:02) (NTS)
ubuntu@ip-172-31-12-33:~/docker-utilities$ sudo docker run --rm myimage bash -c "command -v unzip || echo 'unzip is not installed'"
unzip is not installed
ubuntu@ip-172-31-12-33:~/docker-utilities$
```